CLASS 10 · CBSE · NCERT MATHEMATICS · DPP SHEET

# Chapter 8: Introduction to Trigonometry

Daily practice problems: 10 questions with an answer key

Try every question on your own first. The answer key is on the last page.

1. In $\triangle ABC$, right-angled at $B$, $AB = 24$ cm and $BC = 7$ cm. Find $\sin A$ and $\cos A$.

2. If $\cos A = \frac{4}{5}$, find $\sin A$ and $\tan A$.

3. If $\sec\theta = \frac{13}{12}$, find $\sin\theta$ and $\cot\theta$.

4. Evaluate $\sin^2 30^\circ + \cos^2 60^\circ + \tan^2 45^\circ$.

5. Evaluate $3\tan^2 30^\circ + 4\cos^2 45^\circ - \sec^2 60^\circ$.

6. Evaluate $\dfrac{\sin 60^\circ + \cos 30^\circ}{1 + \tan 45^\circ}$.

7. If $\sin(A+B) = \frac{\sqrt{3}}{2}$ and $\tan(A-B) = \frac{1}{\sqrt{3}}$, with $0^\circ < A + B \le 90^\circ$ and $A > B$, find $A$ and $B$.

8. Prove that $(1 + \cot^2 A)(1 - \cos A)(1 + \cos A) = 1$.

9. Prove that $\cos^4 A - \sin^4 A = 1 - 2\sin^2 A$.

10. Prove that $(\sec A - \cos A)(\cot A + \tan A) = \tan A \sec A$.

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## Answer Key

1. $AC = 25$ cm; $\sin A = \frac{7}{25}$, $\cos A = \frac{24}{25}$
2. $\sin A = \frac{3}{5}$, $\tan A = \frac{3}{4}$
3. $\sin\theta = \frac{5}{13}$, $\cot\theta = \frac{12}{5}$
4. $\frac{1}{4} + \frac{1}{4} + 1 = \frac{3}{2}$
5. $1 + 2 - 4 = -1$
6. $\frac{\sqrt{3}}{2}$
7. $A + B = 60^\circ$, $A - B = 30^\circ$: $A = 45^\circ$, $B = 15^\circ$
8. LHS $= \operatorname{cosec}^2 A \times \sin^2 A = 1$
9. LHS $= (\cos^2 A + \sin^2 A)(\cos^2 A - \sin^2 A) = \cos^2 A - \sin^2 A = 1 - 2\sin^2 A$
10. $\sec A - \cos A = \frac{\sin^2 A}{\cos A}$ and $\cot A + \tan A = \frac{1}{\sin A \cos A}$; the product is $\frac{\sin A}{\cos^2 A} = \tan A \sec A$

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